

THE DISCRETE SCHWARZ-PICK LEMMA FOR CIRCLE PACKINGS REVISITED

ABSTRACT. The Discrete Schwarz-Pick Lemma is a discrete analogue of the classical result from complex analysis, arising from the connection between circle packings and conformal maps established by Thurston. Prior work by Beardon, Stephenson and Van Eeuwen established this lemma for circle packings where circles are tangent or intersect at acute angles, corresponding to inversive distances $I \in (0, 1]$. This paper extends the investigation to circle packings with inversive distances in $(-1, 1]$. Within this regime, we prove that the Discrete Schwarz-Pick Lemma holds for a specific subclass of packings that satisfy a condition on the weights of each triangle. The proof relies on a variational principle for circle packings with inversive distances. In contrast, we also show that the lemma fails for packings containing both tangent ($I = 1$) and disjoint ($I > 1$) circles. We demonstrate this by constructing a specific counterexample on a triangulated disk with four vertices.

1. INTRODUCTION

1.1. Background.

In March of 1985, in the International Symposium in Celebration of the Bieberbach Conjecture, William Thurston suggested the connection between circle packings and conformal maps. William Thurston conjectured that the conformal map from a simply connected domain Ω to the unit disk $\mathbb{D}$ could be approximated by circle packings, a claim famously proven by Rodin and Sullivan in 1987 [1].

Furthermore, in his work constructing hyperbolic structures on 3 manifolds, Thurston introduced the notion of circle packing metrics on triangulated surfaces with non-obtuse intersection angles [2]. Thurston utilized this prescription of angles because the angles of intersecting circles are invariant under Möbius Transformations. For triangulated surfaces with Thurston's construction there are singularities at the vertices. Classic discrete curvature $K(v)$, which is defined as the angle deficit at vertex v , is used to describe the singularity at the vertex.

As a generalization of Thurston's Circle packings, inversive distance circle packings were introduced by Bowers and Stephenson [3]. This generalization allowed for circles of adjacent vertices to be disjoint with the distance between the two circles measured by inversive distance. For more information, see Bowers-Hurdal [4], Stephenson [5], and Guo [6] for more information.

Starting from Thurston's initial idea of approximating conformal maps by discretizing our space with the use of circle packings and the eventual proof by Rodin and Sullivan, work began on discretizing the ideas and theorems of complex analysis and Riemann Surfaces with the use of circle packings. Important theorems such as the Riemann Mapping Theorem and Uniformization got their discretized equivalents [7].

The theorem of interest for this paper is the Discrete Schwarz-Pick Lemma, which is the discretized version of the Schwarz-Pick Lemma in complex analysis. Beardon and Stephenson established the Discrete Schwarz-Pick Lemma for the case of tangent circle packings using discrete Perron's method [8]. Later, Van Eeuwen proved the case for intersection angles φ between two adjacent circles between 0

and $\frac{\pi}{2}$ or inversive distance between 0 and 1 [9]. The difficulty in extending these results to a broader range of inversive distances, lies in the fact that the space of circle packing metrics lacks convexity which breaks many of the arguments needed to establish this result.

This paper relies on an observation by Zhou [10], to establish some bounds on inversive distances to reestablish convexity to the space of circle packing metrics to prove the Discrete Schwarz-Pick Lemma. The main tools are the variational principle established by Colin de Verdiere [11] and expanded by Guo [6], and foundational results on circle packing geometries by Thurston [2]. Additionally, the paper presents a counter example to the Discrete Schwarz-Pick Lemma for the case of $I \geq 1$ (i.e. tangent or disjoint circles) which was conjectured to hold. This is surprising because of the fact that both infinitesimal [6] and global rigidity [12] were proven for this case. Many of the ideas that go into proving global rigidity in this case are also used to prove the Discrete Schwarz-Pick.

1.2. Inversive Distance Circle Packings.

In this subsection, we discuss the concept of inversive distance circle packings introduced by Bowers and Stephenson [3]. For all $n \in \mathbb{N}$, let $[n] := \{1, \dots, n\}$.

Let S be a surface with Euclidean or hyperbolic background geometry and a finite simplicial triangulation $\mathcal{T} = \{V, E, F\}$, where V, E, F represent the vertices, edges, and faces of $\mathcal{T}$. Let $N = |V|$ be a number of vertices. We identify the vertex set V with an index set $[N]$. Thus $E \subset \binom{[N]}{2}$ is the set of edges (pairs of indices) and $F \subset \binom{[N]}{3}$ is the set of faces. We will adopt the following notation convention for convenience:

1. For generic indices represented by variables (e.g. $i, j, k \in [N]$) we will denote an edge simply as $ij \in E$ and a face as $ijk \in F$.
2. For specific numerical indices, we will use formal set notation (e.g. an edge $\{1, 2\} \in E$ or a face $\{1, 2, 3\} \in F$).

Let $I : E \rightarrow (-1, \infty)$ be a function that assigns each edge $\{i, j\}$ a inversive distance $I_{ij} \in (-1, \infty)$. The triple $(S, \mathcal{T}, I)$ will be referred to as a **weighted triangulated surface**.

For all functions $f : V \rightarrow \mathbb{R}$, let $f_i := f(i)$ and thus f can be viewed as the vector $(f_1, \dots, f_N) \in \mathbb{R}^N$. Let $\mathbb{R}_{>0} := (0, \infty)$. Any map $r : V \rightarrow \mathbb{R}_{>0}$ is called a **radius function**. The radius function on a weighted triangulated surface induces a edge length map $l_r : E \rightarrow \mathbb{R}_{>0}$ where

$$l_r(ij) = \sqrt{r_i^2 + r_j^2 + 2r_i r_j I_{ij}} \quad (1)$$

if the background geometry is Euclidean and

$$l_r(ij) = \cosh^{-1}(\cosh(r_i) \cosh(r_j) + I_{ij} \sinh(r_i) \sinh(r_j)) \quad (2)$$

if the background geometry is hyperbolic. Note that the length $l_r(ij)$ in (1) and (2) is well defined if $r_i, r_j > 0$ and $I_{ij} > -1$. Furthermore the inversive distance I_{ij} determines the geometry of circles attached to v_i and v_j respectively :

- If $I_{ij} \in (-1, 1]$ the circles corresponding to vertices i and j respectively intersect. There exists a unique angle $\Phi_{ij} \in [0, \pi)$ such that $I_{ij} = \cos(\Phi_{ij})$. Φ_{ij} is the angle formed by the tangent lines of the circles at the intersection

point. See Figure 1 for reference. The intersection is obtuse if $I_{ij} \in (-1, 0)$, orthogonal if $I_{ij} = 0$, and acute if $I_{ij} \in (0, 1)$.

- If $I_{ij} = 1$, the circles are tangent.
- If $I_{ij} > 1$, the circles are disjoint.

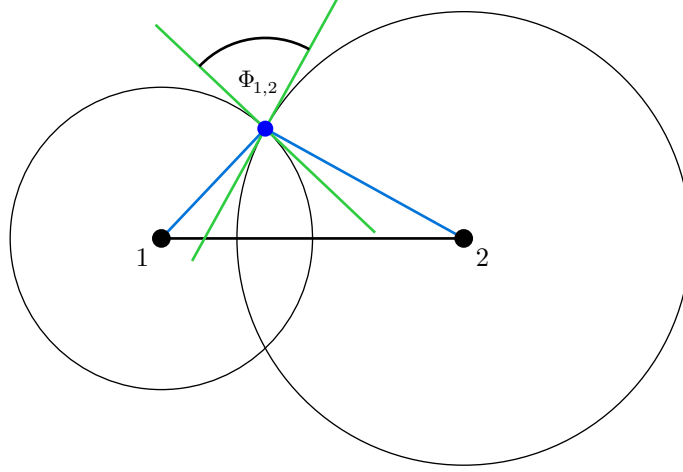


FIGURE 1. Intersection Angle in Circle Packings

If for each face $\{i, j, k\} \in F$ the induced lengths l_r satisfy the triangle inequalities

$$\begin{aligned} l_r(ij) + l_r(jk) &> l_r(ik) \\ l_r(ik) + l_r(jk) &> l_r(ij) \\ l_r(ij) + l_r(ik) &> l_r(jk) \end{aligned} \tag{3}$$

then r is called a **circle packing metric**. Such a metric induces a combinatorial curvature on the surface $K_r := V \rightarrow (-\infty, 2\pi)$ where

$$K_r(i) := 2\pi - \sum_{\alpha > i} \alpha \tag{4}$$

where $\alpha > v$ means all the angles incident to i . Note that $r \rightarrow K_r$ is a continuous function.

1.3. Plan of the Paper.

The paper is organized as follows. In Section 2 we prove the Discrete Schwarz-Pick Lemma for Circle Packings for a broader range of intersection behaviors for hyperbolic and Euclidean background. In Section 3 we present a counterexample to the Discrete Schwarz-Pick Conjecture for the case of tangent and disjoint circles.

2. DISCRETE SCHWARZ-PICK LEMMA FOR CIRCLE PACKING WITH INTERSECTIONS

2.1. Triangle Preliminaries in Euclidean background geometry.

Let $(S, \mathcal{T}, I)$ be a weighted triangulated surface with Euclidean background geometry and let r be a radius function, and l_r and K_r be the induced edge lengths and discrete curvatures respectively. Suppose $\tau = ijk \in F$. For this section we will

temporarily adopt standard geometric conventions where $l_i = l_r(jk)$ and $I_i = I(jk)$ are the edge length and inversive distance of the edge opposite vertex i . Let

$$\gamma_\tau = (I_i + I_j I_k, I_j + I_i I_k, I_k + I_i I_j) = (\gamma_\tau^i, \gamma_\tau^j, \gamma_\tau^k). \quad (5)$$

As noted in [6], the edge lengths l_i, l_j, l_k satisfy the triangle inequalities if and only if

$$\begin{aligned} 0 &< (l_i + l_j + l_k)(l_j + l_k - l_i)(l_i + l_k - l_j)(l_i + l_j - l_k) \\ &= \left((l_j + l_k)^2 - l_i^2\right)(l_i - (l_j - l_k))(l_i + (l_j - l_k)) \\ &= \left((l_j + l_k)^2 - l_i^2\right)(l_i^2 - (l_j - l_k)^2) \\ &= (-l_i^2 + l_j^2 + 2l_j l_k + l_k^2)(l_i^2 - l_j^2 + 2l_j l_k - l_k^2) \\ &= -l_i^4 - l_j^4 - l_k^4 + 2l_i^2 l_j^2 + 2l_i^2 l_k^2 + 2l_j^2 l_k^2 \end{aligned} \quad (6)$$

Substituting the definition for edge lengths defined in (1), by direct computation we have

$$\begin{aligned} &\frac{1}{4}(l_i + l_j + l_k)(l_j + l_k - l_i)(l_i + l_k - l_j)(l_i + l_j - l_k) \\ &= r_i^2 r_j^2 (1 - I_k^2) + r_i r_k (1 - I_j^2) + r_j^2 r_k^2 (1 - I_i^2) \\ &\quad + 2r_i r_j r_k (r_i \gamma_\tau^i + r_j \gamma_\tau^j + r_k \gamma_\tau^k) \end{aligned} \quad (7)$$

Thus we have the following result on the Euclidean triangle inequalities.

Lemma 2.1.1 ([13], Lemma 2.1) Suppose $(S, \mathcal{T}, I)$ is a weighted triangulated surface, r is a radius function, and $\tau = \triangle v_i v_j v_k$ is a topological triangle in F . Then the edge lengths defined in (1) satisfy the triangle inequalities if and only if

$$\begin{aligned} &r_i^2 r_j^2 (1 - I_k^2) + r_i r_k (1 - I_j^2) + r_j^2 r_k^2 (1 - I_i^2) \\ &\quad + 2r_i r_j r_k (r_i \gamma_\tau^i + r_j \gamma_\tau^j + r_k \gamma_\tau^k) > 0 \end{aligned} \quad (8)$$

The following corollary obtained in [10] directly follows from Lemma 2.1.1.

Corollary 2.1.2 If $I_i, I_j, I_k \in (-1, 1]$ and $\gamma_\tau \in [0, \infty)^3$, then the triangle inequalities are satisfied for any $(r_i, r_j, r_k) \in \mathbb{R}_{>0}^3$.

It immediately follows that if for all $\tau \in F$ we have that $\gamma_\tau \in \mathbb{R}_{\geq 0}^3$, then the space of circle packing metrics on $(S, \mathcal{T}, I)$ is $\mathbb{R}_{>0}^N$.

2.2. Infinitesimal Rigidity of Euclidean Inversive Distance Circle Packings.

Let $(S, \mathcal{T}, I)$ be a weighted triangulated surface with euclidean background where $-1 < I \leq 1$. Let $\tau = ijk \in F$ where $\gamma_\tau \in \mathbb{R}_{\geq 0}^3$. By Corollary 2.1.2, $\forall (r_i, r_j, r_k) \in \mathbb{R}_{>0}^3$ induces edge lengths which form a triangle. Let $(u_i, u_j, u_k) = (\ln(r_i), \ln(r_j), \ln(r_k))$. For all $a \in \{i, j, k\}$, let θ_a inner angle of τ at the vertex a . We have the following useful lemma regarding the monotonicity of θ_a with respect to (u_i, u_j, u_k) , and thus (r_i, r_j, r_k) .

Lemma 2.2.3 ([13]) For all $\tau = ijk \in F$ where $\gamma_\tau \in \mathbb{R}_{\geq 0}^3$, we have the following:

1. $\frac{\partial \theta_a}{\partial u_b} = \frac{\partial \theta_b}{\partial u_a} > 0$ for $a, b \in \{i, j, k\}$ where $a \neq b$.
2. $\frac{\partial \theta_a}{\partial u_a} < 0$ for $a \in \{i, j, k\}$.
3. The jacobian $J_u(\theta)$ is symmetric and negative semi definite, with null space $\mathbb{R}[1, 1, 1] = \{[t, t, t] \mid t \in \mathbb{R}\}$.
4. The differential 1-form $\sum_{a \in \{i, j, k\}} \theta_a du_a$ is closed on $\mathbb{R}^3$ (and hence exact) and

$$W_\tau(u) = \int_0^u \sum_{a \in \{i, j, k\}} \theta_a du_a \quad (10)$$

is a well defined concave function of $u \in \mathbb{R}^3$ satisfying

$$\nabla W_\tau = (\theta_i, \theta_j, \theta_k) \quad (11)$$

Furthermore, W_τ is strictly concave when restricted to $\{tu + (1-t)\hat{u} \mid t \in \mathbb{R}\}$, provided that $u - \hat{u} \neq (a, a, a)$ for $a \in \mathbb{R}$.

Proof Part 1 of the lemma comes from Lemma 2.5 in Xu [13]. Part 2 holds because $\theta_i + \theta_j + \theta_k = \pi$. Thus

$$\frac{\partial \theta_i}{\partial u_i} + \frac{\partial \theta_j}{\partial u_i} + \frac{\partial \theta_k}{\partial u_i} = 0. \quad (12)$$

Since $\frac{\partial \theta_a}{\partial u_b} > 0$ for $a \neq b$, by part 1 we deduce that $\frac{\partial \theta_i}{\partial u_i} < 0$. Part 3 and 4 hold because of Lemma 2.6 and Lemma 2.10 respectively from Xu [13]. $\square$

2.3. Triangle Preliminaries in hyperbolic background geometry.

Let $(S, \mathcal{T}, I)$ be a weighted triangulated surface with hyperbolic background geometry and let r be a radius function, and l_r and K_r be the induced edge lengths and discrete curvatures respectively. Suppose $\tau = ijk \in F$, for this section we will temporarily adopt standard geometric conventions where $l_i = l_r(jk)$ and $I_i = I(jk)$ are the edge length and inversive distance of the edge opposite vertex i . Let

$$\gamma_\tau = (I_i + I_j I_k, I_j + I_i I_k, I_k + I_i I_j) = (\gamma_\tau^i, \gamma_\tau^j, \gamma_\tau^k). \quad (13)$$

Let

$$S_i = \sinh(r_i) \text{ and } C_i = \cosh(r_i). \quad (14)$$

Working the algebra to a inequality similar to (6), we get the following lemma regarding the circle packing metrics for our triangle τ .

Lemma 2.3.4 ([6], Lemma 3.1) Suppose $(S, \mathcal{T}, I)$ is a weighted triangulated surface with hyperbolic background geometry. Let r be a radius function and $\tau = ijk \in F$. Then the edge lengths defined in (1) satisfy the triangle inequalities if and only if

$$\begin{aligned} & 2S_i^2 S_j^2 S_k^2 (1 + I_i I_j I_k) + S_i^2 S_j^2 (1 - I_k^2) + S_i^2 S_k^2 (1 - I_j^2) + S_j^2 S_k^2 (1 - I_i^2) \\ & + 2S_i S_j S_k (S_i C_j C_k \gamma_\tau^i + C_i S_j C_k \gamma_\tau^j + C_i C_j S_k \gamma_\tau^k) > 0 \end{aligned} \quad (15)$$

The following corollary obtained in [10] directly follows from Lemma 2.3.4.

Corollary 2.3.5 ([13]) If $I_i, I_j, I_k \in (-1, 1]$ and $\gamma_\tau \in [0, \infty)^3$, then the triangle inequalities are satisfied for any $(r_i, r_j, r_k) \in \mathbb{R}_{>0}^3$. Thus $\Omega_\tau = \mathbb{R}_{>0}^3$.

2.4. Infinitesimal Rigidity of Hyperbolic Inversive Distance Circle Packings.

Let $(S, \mathcal{T}, I)$ be a weighted triangulated surface with hyperbolic background geometry where $-1 < I \leq 1$. Let $\tau = ijk \in F$ where $\gamma^\tau \in \mathbb{R}_{\geq 0}^3$. By Corollary 2.3.5, $\forall (r_i, r_j, r_k) \in \mathbb{R}_{>0}^3$ induces edge lengths which form a triangle. Let

$$u = (u_i, u_j, u_k) = \left(\ln\left(\tanh\left(\frac{r_i}{2}\right)\right), \ln\left(\tanh\left(\frac{r_j}{2}\right)\right), \ln\left(\tanh\left(\frac{r_k}{2}\right)\right) \right) \quad (16)$$

. Let θ_a inner angle of τ at the vertex v_a for $a \in \{i, j, k\}$. We have the following useful lemma regarding the monotonicity of θ_a with respect to (u_i, u_j, u_k) , and thus (r_i, r_j, r_k) .

Lemma 2.4.6 ([13]) For all $\tau = ijk \in F$ where $\gamma_\tau \in \mathbb{R}_{\geq 0}^3$, we have the following:

1. $\frac{\partial \theta_a}{\partial u_b} = \frac{\partial \theta_b}{\partial u_a} > 0$ for $a, b \in \{i, j, k\}$ where $a \neq b$.
2. $\frac{\partial \theta_a}{\partial u_a} < 0$ for $a \in \{i, j, k\}$.
3. The jacobian $J_u(\theta)$ is symmetric and negative semi definite, with null space

$$\mathbb{R}[1, 1, 1] = \{[t, t, t] \mid t \in \mathbb{R}\}. \quad (17)$$

4. The differential 1-form $\sum_{a \in \{i, j, k\}} \theta_a du_a$ is closed on $\mathbb{R}^3$ (and hence exact) and

$$W_\tau(u) = \int_0^u \sum_{a \in \{i, j, k\}} \theta_a du_a \quad (18)$$

is a well defined concave function of $u \in \mathbb{R}^3$ satisfying

$$\nabla W_\tau = (\theta_i, \theta_j, \theta_k) \quad (19)$$

Furthermore, W_τ is strictly concave when restricted to $\{tu + (1-t)\hat{u} \mid t \in \mathbb{R}\}$, provided that $u - \hat{u} \neq (a, a, a)$ for $a \in \mathbb{R}$.

Proof Part 1 of the lemma holds by Lemma 3.6 in Xu [13]. By Remark 11 in Xu [13],

$$\frac{\partial \theta_i}{\partial u_i} + \frac{\partial \theta_j}{\partial u_i} + \frac{\partial \theta_k}{\partial u_i} < 0 \quad (20)$$

since $\frac{\partial \theta_a}{\partial u_b} > 0$ for $a \neq b$ by 1, $\frac{\partial \theta_a}{\partial u_a} < 0$, thus Part 2 holds. Part 3 holds by Lemma 3.7 in Xu [13]. Finally, Part 3 holds by Lemma 3.9 in Xu [13]. $\square$

2.5. Degeneration of Circle Packing Metrics.

Let $(S, \mathcal{T}, I)$ be a closed connected weighted triangulated surface with Euclidean or hyperbolic ground geometry where $-1 < I \leq 1$. Let $\Phi : E \rightarrow [0, \pi)$ be the corresponding intersection angle for I , ie $I(e) = \cos(\Phi(e))$. $\forall J \subset V$, let F_J be the subcomplex of $\mathcal{T}$ consisting of the 2-simplices whose vertices are in J and

$$\text{Lk}(J) = \{(\tau, v) \in F \times J \mid \{v\} = \tau \cap J\}. \quad (21)$$

Lemma 2.5.7 Let J be some subset of V . Let $(r_n)_{n \in \mathbb{N}}$ be a sequence of circle packing metrics for $(S, \mathcal{T}, I)$ ($-1 < I \leq 1$) where if $j \in J$

$$\lim_{n \rightarrow \infty} r_n(j) = 0 \quad (22)$$

and if $k \notin J$,

$$\lim_{n \rightarrow \infty} r_n(k) > 0. \quad (23)$$

Then

$$\lim_{n \rightarrow \infty} \sum_{j \in J} K_{r_n}(j) = 2\pi\chi(F_J) - \sum_{(\tau, v) \in \text{Lk}(J)} (\pi - \Phi(\tau \setminus \{v\})) \quad (24)$$

Proof This proof follows the argument outlined in Proposition 4.1 in Chow and Luo [14]. $\forall \tau \in F$ and $\forall i \in \tau$, θ_i^τ be the inner angle of i in triangle τ . Let T_1 , T_2 , and T_3 be the sets of triangles with one, two, and three vertices in J . Thus for any circle packing metric r , we have following:

$$\begin{aligned} \sum_{j \in J} K_r(j) &= 2\pi|J| - \sum_{j \in J} \sum_{\substack{\tau \in F \\ j \in \tau}} \theta_j^\tau \\ &= 2\pi|J| - \sum_{\tau \in F} \sum_{j \in \tau \cap J} \theta_j^\tau \\ &= 2\pi|J| - \left(\sum_{\substack{\tau \in T_1 \\ i \in \tau \cap J}} \theta_i^\tau + \sum_{\substack{\tau \in T_2 \\ i, j \in \tau \cap J \text{ distinct}}} \theta_i^\tau + \theta_j^\tau + \sum_{\substack{\tau \in T_3 \\ \tau = ijk}} \theta_i^\tau + \theta_j^\tau + \theta_k^\tau \right) \end{aligned} \quad (25)$$

Note there is bijection between T_1 and $\text{Lk}(J)$, so we have the following equivalence:

$$\sum_{\substack{\tau \in T_1 \\ i \in \tau \cap J}} \theta_i^\tau = \sum_{(i, \tau) \in \text{Lk}(J)} \theta_i^\tau \quad (26)$$

Given that $\forall i \in J$, $r_n(i) \rightarrow 0$ we know the following:

1. $\forall \tau = ijk \in T_3$ as $(r_n(i), r_n(j), r_n(k)) \rightarrow (0, 0, 0)$, ijk approaches a Euclidean triangle. Thus $\theta_i^\tau + \theta_j^\tau + \theta_k^\tau \rightarrow \pi$.
2. $\forall \tau \in T_2$ and $i, j \in \tau \cap J$ distinct as $(r_n(i), r_n(j)) \rightarrow (0, 0)$, ijk approaches a geodesic segment. Thus $\theta_k^\tau \rightarrow 0 \Rightarrow \theta_i^\tau + \theta_j^\tau \rightarrow \pi$.
3. $\forall \tau = ijk \in T_1$ and $i \in \tau \cap J$, then i approaches the intersection point of the circles corresponding to j and k . Thus $\theta_i^\tau \rightarrow \pi - \Phi(jk) = \pi - \Phi(\tau \setminus \{i\})$.

Putting all that together, we see:

$$\begin{aligned}
\lim_{n \rightarrow \infty} \sum_{j \in J} K_{r_n}(j) &= 2\pi|J| - \left(\sum_{(i, \tau) \in \text{Lk}(J)} \pi - \Phi(\tau \setminus \{i\}) + |T_2|\pi + |T_3|\pi \right) \\
&= 2\pi \left(|J| - \frac{1}{2}|T_2| - \frac{1}{2}|T_3| \right) - \sum_{(i, \tau) \in \text{Lk}(J)} \pi - \Phi(\tau \setminus \{i\}) \quad (27) \\
&= 2\pi \left(|J| - \frac{1}{2}(|T_2| + 3|T_3|) + |T_3| \right) - \sum_{(i, \tau) \in \text{Lk}(J)} \pi - \Phi(\tau \setminus \{i\})
\end{aligned}$$

$|J|$ is the number of vertices in F_J . $|T_3|$ is the number of 2-simplices in F_J . By construction, the number of edges is $\frac{1}{2}(|T_2| + 3|T_3|)$. Thus

$$\lim_{n \rightarrow \infty} \sum_{j \in J} K_{r_n}(j) = 2\pi\chi(F_J) - \sum_{(\tau, v) \in \text{Lk}(J)} (\pi - \Phi(\tau \setminus \{v\})) \quad (28)$$

□

We get the following corollary from the lemma above.

Corollary 2.5.8 If $\forall \tau \in F$ we have that $\gamma^\tau \in \mathbb{R}_{\geq 0}^3$, then $\forall r : V \rightarrow \mathbb{R}_{>0}^3$ circle packing metrics

$$\sum_{j \in J} K_r(v_j) > 2\pi\chi(F_J) - \sum_{(\tau, v) \in \text{Lk}(J)} (\pi - \Phi(\tau \setminus \{v\})) \quad (29)$$

Proof Since the background geometry is Euclidean or Hyperbolic, we know that the maximum sum of interior angles of a triangle is π . Thus for any circle packing metric r ,

$$\sum_{\substack{\tau \in A_2 \\ i, j \in \tau \cap J \text{ distinct}}} \theta_i^\tau + \theta_j^\tau < \pi|T_2| \text{ and } \sum_{\substack{\tau \in A_3 \\ i, j, k \in \tau \cap J \text{ distinct}}} \theta_i^\tau + \theta_j^\tau + \theta_k^\tau < \pi|T_3|. \quad (30)$$

Furthermore since $\forall \tau \in F$ we have that $\gamma^\tau \in \mathbb{R}_{>0}$, we can apply [Lemma 2.2.3](#) and [Lemma 2.4.6](#) to get monotonicity of interior angles. $\forall \tau \in A_1$ where $i \in \tau \cap J$, we know that θ_i^τ is monotonically increasing, thus $\pi - \Phi(\tau \setminus \{v\})$ is the upper bound for θ_i^τ . Thus we have:

$$\sum_{j \in J} K_r(j) > 2\pi|J| - \left(\sum_{(i, \tau) \in \text{Lk}(J)} \pi - \Phi(\tau \setminus \{i\}) + |T_2|\pi + |T_3|\pi \right) \quad (31)$$

Proceeding similarly to [Lemma 2.5.7](#), we see that

$$\sum_{j \in J} K_r(v_j) > 2\pi\chi(F_J) - \sum_{(\tau, v) \in \text{Lk}(J)} (\pi - \Phi(\tau \setminus \{v\})) \quad (32)$$

□

2.6. Main Theorem.

Theorem 2.6.9 Let $(S, \mathcal{T}, I)$ be a closed connected weighted triangulated surface where $I \in (-1, 1]$. Let Φ be its corresponding intersection angle function. Let $A, B \subset [N]$ where $N = A \sqcup B$ and $A \neq \emptyset$ and in the case Euclidean background,

$B \neq \emptyset$. If $\forall \tau \in F$ we have that $\gamma^\tau \in \mathbb{R}_{\geq 0}^3$, then for any circle packing metrics R and r for $(S, \mathcal{T}, I)$ where $R|_B \geq r|_B$ and $K_R|_A \geq K_r|_A$ it means that $R \geq r$.

Proof

Let

$$X := \{x \in \mathbb{R}_{\geq 0}^N \mid x|_B \geq x|_B \text{ and } K_x|_A \geq K_r|_A\}. \quad (33)$$

Note that $r, R \in X$ by assumption. We want to show that $\forall v \in V$,

$$r(v) = \inf\{x(v) : x \in X\}. \quad (34)$$

This will imply that $R \geq r$ and finish the proof.

$\forall a, b \in X$, let $c_i = \min(a_i, b_i)$, thus $c = \min(a, b) := (c_1, \dots, c_N)$. Clearly, $c_j \geq r_j$ for all $j \in B$. $\forall v \in V_A$, let $\{u_1, \dots, u_m\}$ the set of vertices adjacent to v . Assume without a loss of generality that, $c(v) = a(v)$. By assumption $c(u_i) \leq a(u_i)$, thus by part 1 of [Lemma 2.2.3](#) and part 1 of [Lemma 2.4.6](#), $K_c(v) \geq K_a(v)$. Thus $K_c(v) \geq K_r(v)$. Thus $c \in X$ and X is closed under minimums.

Let $w = \inf X$ (component-wise). $\forall v \in V$, let $(x_n^v)_{n \in \mathbb{N}}$ be a sequence in X such that

$$w(v) = \lim_{n \rightarrow \infty} x_n^v(v). \quad (35)$$

Let $(y_n)_{n \in \mathbb{N}}$ be a sequence where

$$y_n := \min\{x_n^v \mid v \in V\}. \quad (36)$$

By closure under minimums, $y_n \in X$ for all $n \in \mathbb{N}$. Since $\forall n \in \mathbb{N}$ and $v \in V$ $y_n \leq x_n^v$, that means

$$\limsup_{n \rightarrow \infty} y_n(v) \leq \limsup_{n \rightarrow \infty} x_n^v(v) = \lim_{n \rightarrow \infty} x_n^v(v) = w(v). \quad (37)$$

At the same time since w is the componentwise infimum that means that $\forall n \in \mathbb{N}$ $y_n(v) \geq w(v)$ which means

$$w(v) \leq \liminf_{n \rightarrow \infty} y_n(v) \Rightarrow \lim_{n \rightarrow \infty} y_n(v) = w(v). \quad (38)$$

We will use this sequence y_n to verify that $w \in X$. We have to check that $w \in \mathbb{R}_{\geq 0}^N$, $w|_B \geq r|_B$, and $K_w|_A \geq K_r|_A$.

Assume the contrary that $w \notin \mathbb{R}_{\geq 0}^N$ thus $\exists v \in V$ such that $w(v) = 0$. Let $J := \{j \in [N] \mid w(v_j) = 0\}$. By [Lemma 2.5.7](#), we have that

$$\lim_{n \rightarrow \infty} \sum_{i \in J} K_{y_n}(v_i) = 2\pi\chi(F_J) - \sum_{(\tau, v) \in \text{Lk}(J)} (\pi - \Phi(\tau \setminus \{v\})). \quad (39)$$

Note that $\forall v \in B$, $w(v) \geq r(v) > 0 \Rightarrow J \cap B = \emptyset$ which means $J \subset A$. Thus $\forall j \in J$,

$$K_{y_n}(v_j) \geq K_r(v_j) \quad (40)$$

which means

$$\lim_{n \rightarrow \infty} \sum_{j \in J} K_{y_n}(v_j) \geq \sum_{j \in J} K_r(v_j). \quad (41)$$

Applying [Corollary 2.5.8](#) for $\sum_{j \in J} K_r(v_j)$ we see that

$$\lim_{n \rightarrow \infty} \sum_{j \in J} K_{y_n}(v_j) \geq \sum_{j \in J} K_r(v_j) > 2\pi\chi(F_J) - \sum_{(\tau, v) \in \text{Lk}(J)} (\pi - \Phi(\tau \setminus \{v\})) \quad (42)$$

which is a contradiction. Thus by contradiction $w \in \mathbb{R}_{>0}^N$ and w is a circle packing metric for $(S, \mathcal{T}, I)$.

$\forall n \in \mathbb{N}$, we know that $y_n|_B \geq r|_B$ and

$$w = \lim_{n \rightarrow \infty} y_n \quad (43)$$

which implies that $w|_B \geq r|_B$. By the definition of infimum, $r|_B \geq w|_B$. Thus $r|_B = w|_B$.

Since $r \rightarrow K_r$ is a continuous function, $\forall i \in A$,

$$K_w(v_i) = \lim_{n \rightarrow \infty} K_{y_n}(v_i) \quad (44)$$

by the sequential definition of continuity. $K_{y_n}(v_i) \geq K_r(v_i) \Rightarrow K_w(v_i) \geq K_r(v_i)$. Thus $w = X$.

We know that $w = X$ and $w|_B = r|_B$, we just need to show that $w|_A = r|_A$. Assume the contrary that $\exists i \in A$ such that $K_w(v_i) > K_r(v_i)$. $\forall t \in (0, w(v_i))$, let $s_t : V \rightarrow \mathbb{R}_{>0}^+$ where

$$s_t(v) = \begin{cases} w(v) - t & \text{if } v = v_i \\ w(v) & \text{otherwise} \end{cases} \quad (45)$$

Since $i \in A$, $s_t|_B = w|_B \geq x|_B$. Let $U \subset V$ be the subset of adjacent vertices to i . Since $s_t|_U = w|_U$ and $s_t(i) < w(i)$. By Part 1 of both **Lemma 2.2.3** and **Lemma 2.4.6**, we $K_{s_t}(i) < K_w(i)$ and $K_{s_t}(j) > K_w(j) \geq K_r(j)$ for all $j \in U$. By continuity, $\exists t_0 \in (0, w(i))$ such that $K_r(i) \leq K_{s_{t_0}}(i) < K_w(i)$. Thus $s_{t_0} \in X$, however note that $s_{t_0}(i) < w(i)$. This is a contradiction since w is a componentwise infimum. Thus by contradiction $K_w|_A = K_r|_A$.

Now we have to show that $r = w$ using the facts that $r|_B = w|_B$ and $K_r|_A = K_w|_A$. Let

$$S = \begin{cases} \mathbb{R} & \text{if background geometry is Euclidean} \\ (0, \infty) & \text{if background geometry is hyperbolic} \end{cases} \quad (46)$$

Let $\varphi : (0, \infty) \rightarrow S$ where

$$\varphi(x) := \begin{cases} \ln(x) & \text{if background geometry is Euclidean} \\ \ln(\tanh(\frac{x}{2})) & \text{if background geometry is hyperbolic} \end{cases} \quad (47)$$

φ is a homeomorphism so φ^{-1} exists and is a continuous. Let $\mathcal{R} := \{x \in \mathbb{R}_{>0}^N : \varphi^{-1}|_B = r|_B\}$. Let $W : S^N \rightarrow \mathbb{R}$ where

$$W(x) := \sum_{\tau = ijk \in F} W_\tau(x_i, x_j, x_k) \quad (48)$$

where W_τ is defined in **Lemma 2.2.3** and **Lemma 2.4.6**. $\forall i \in V$,

$$\begin{aligned}
 \frac{\partial W}{\partial x_a} &= \sum_{\tau=ijk \in F} \frac{\partial W_\tau}{\partial x_a} \\
 &= \sum_{\tau=ajk \in F} \frac{\partial W_\tau}{\partial x_a} \\
 &= \sum_{\tau=ajk \in F} \theta_a^\tau \\
 &= 2\pi - K_{\varphi^{-1}(x)}(a),
 \end{aligned} \tag{49}$$

where the second to last equality comes from Part 2 of [Lemma 2.2.3](#) and [Lemma 2.4.6](#) and the last equality comes from the definition of discrete curvature. Thus

$$\nabla W(x) = 2\pi - K_{\varphi^{-1}(x)}. \tag{50}$$

Let $F = W|_{\mathcal{R}}$, thus $F : \mathbb{R}_{>0}^A \rightarrow \mathbb{R}$ and

$$\nabla F(x) = 2\pi - K_{\varphi^{-1}(x)}|_A. \tag{51}$$

$\forall x, y \in \mathcal{R}$ such that $xx \neq y$, take $M = \{i \in V \mid x(i) \neq y(i)\}$ and $N = \{i \in V \mid x(i) = y(i)\}$. Note $M \neq \emptyset$, since $x \neq y$, $B \subset N$ and $M \cap N = \emptyset$. Since $(S, \mathcal{T})$ is connected, then the graph of the triangulation is connected. Thus $\exists(i, j) \in M \times N$ such that $\exists k \in V$ such that $ijk \in F$. By definition

$$(x(i) - y(i), x(j) - y(j), x(k) - y(k)) \neq (0, 0, 0) \tag{52}$$

because $x(i) \neq y(i)$. Thus W_{ijk} is strictly concave on $\{tu + (1+t)\hat{u}\}$ where $u = (x(i), x(j), y(k))$ and $\hat{u} = (y(i), y(j), y(k))$. Thus W_τ is strictly concave on $\mathcal{R}$, thus W and F are strictly concave on $\mathcal{R}$. Which implies that ∇F is injective and $x \rightarrow K_{\varphi^{-1}(x)}|_A$ is injective. Thus since $K_w|_A = K_r|_A \Rightarrow w|_A = r|_A \Rightarrow w = r$. Thus $r = \inf(X)$ and $R \geq r$. $\square$

3. THE DISCRETE SCHWARZ-PICK LEMMA FOR $I \geq 1$

For $I \geq 1$, it was conjectured that the Discrete Schwarz-Pick Lemma would hold for all $I \geq 1$, because for such circle packings it is known that rigidity holds, for hyperbolic (respectively Euclidean) geometry discrete curvature determines (respectively up to scale) the circle packing metric, by work done by Luo [12]. In such circle packings, circles corresponding to adjacent vertices are allowed to be disjoint or tangent.

Conjecture 3.10 Let $(S, \mathcal{T}, I)$ be a closed connected weighted triangulated surface in Euclidean or hyperbolic geometry such that $I \geq 1$. Let $V = A \sqcup B$ such that $A \neq \emptyset$ and $B \neq \emptyset$, if the background geometry is Euclidean. Then for all circle packing metrics R and r on $(S, \mathcal{T}, I)$ where $R|_B \geq r|_B$ and $K_R|_A \geq K_r|_A$, $R \geq r$.

3.1. Counterexample for the Conjecture.

The statement of the conjecture concerns closed surface, but a counterexample on a triangulated disk is sufficient. By a standard doubling argument, any such counterexample can be extended to a closed surface.

Let $(S, \mathcal{T}, I)$ be a disk with boundary with interior vertices A and boundary vertices B . Construct a closed surface S' by taking two copies of S and identifying their boundaries with the identity map. This induces a triangulation $\mathcal{T}'$ and inversive distance I' on S' . Given to circle packing metrics r and R on $(S, \mathcal{T}, I)$, we can define corresponding circle packing metrics r' and R' on $(S', \mathcal{T}', I')$. If the premises of the conjecture (ie $r|_B \leq R|_B$ and $K_r|_A \leq K_R|_A$) hold on the disk, they will hold on the doubled surface. Therefore if we find a interior vertex $i \in A$ such that $r(i) > R(i)$, this violation will carry over to the doubled surface and thus refute the conjecture for the closed surface.

Conjecture 3.1.11 Let $(S, \mathcal{T}, I)$ be a connected weighted triangulated disk in Euclidean geometry such that $I \geq 1$. Let A be the interior vertices and B be the boundary vertices. Then for all circle packing metrics R and r on $(S, \mathcal{T}, I)$ where $R|_B \geq r|_B$ and $K_R|_A \geq K_r|_A$, $R \geq r$.

Thus if **Conjecture 3.1.10** holds for Euclidean background geometry then **Conjecture 3.1.11** holds for Euclidean background geometry. Additionally if we find a counterexample for **Conjecture 3.1.11**, then we have a counterexample for **Conjecture 3.1.10** for Euclidean background geometry.

3.2. Construction of the Counterexample.

We will now construct a weighted triangulated surface with Euclidean background geometry with $I \geq 1$ which can be doubled to create a counterexample for the conjecture. Let $V = [4]$, $A = \{4\}$, and $B = [3]$. Thus $A \sqcup B = V$. Let the edges be $E = \{\{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 4\}, \{2, 4\}, \{3, 4\}\}$ and let the faces be $F = \{\{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}\}$. Let $I : E \rightarrow [1, \infty]$ where

$$I(e) := \begin{cases} 4 & \text{if } e = (2, 4) \\ 3 & \text{if } e = (3, 4) \\ 1 & \text{otherwise} \end{cases} \quad (53)$$

Thus $(\mathbb{D}, \{V, E, F\}, I)$ is a weighted triangulated disk.

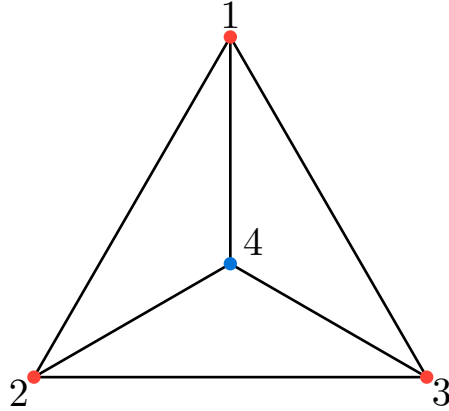


FIGURE 2. Triangulation of Counterexample

Let $r, R : V \rightarrow \mathbb{R}_{>0}$, where

$$r_E = (100, 100, 100, 155) \text{ and } R_E = (110, 240, 220, 150). \quad (54)$$

Note we are using the vector definition of functions from V to $\mathbb{R}$.

3.3. Verification of Counterexample.

3.3.1. Verification of Circle Packing Metrics.

To verify that such construction forms a valid counterexample, we first need to verify that r, R are valid circle packing metrics. To do this we have to calculate the edge lengths using (1). The induced edge lengths for r is

Edge e	Edge Length $l_r(e)$
$\{1, 2\}$	200.0
$\{1, 3\}$	200.0
$\{2, 3\}$	200.0
$\{1, 4\}$	255.0
$\{2, 4\}$	$\sqrt{100^2 + 155^2 + 2(155)(100)(4)} \approx 397.52358$
$\{3, 4\}$	$\sqrt{100^2 + 155^2 + 2(155)(100)(3)} \approx 356.40567$

and the induced edge lengths for R

Edge e	Edge Length $l_R(e)$
$\{1, 2\}$	350.0
$\{1, 3\}$	330.0
$\{2, 3\}$	460.0
$\{1, 4\}$	260.0
$\{2, 4\}$	$\sqrt{240^2 + 150^2 + 2(150)(240)(4)} \approx 606.71245$
$\{3, 4\}$	$\sqrt{220^2 + 150^2 + 2(220)(150)(3)} \approx 518.55569$

We have to use the edge lengths to verify the triangle inequality. For side lengths l_1, l_2 , and l_3 , to see if the lengths l_1, l_2 , and l_3 form a triangle it suffices to see if they satisfy the following inequality:

$$(l_1 + l_2 + l_3)(l_1 + l_2 - l_3)(l_1 + l_3 - l_2)(l_2 + l_3 - l_1) > 0 \quad (55)$$

If we plug in the values for l_r and l_R for each triangle we can see they satisfy the triangle inequality. Thus r and R are valid circle packing metrics.

3.4. Computation of Discrete Curvature.

To compute the induced discrete curvature at a vertex, we first need to compute the inner angle of each triangle the vertex is a part of. To compute the inner angle at a vertex a , we can use the Euclidean cosine law. Let A be the length of edge opposite to a and B and C the length of the other edges.

$$\cos(\theta_a) = \frac{B^2 + C^2 - A^2}{2BC} \Rightarrow \theta_a = \arccos\left(\frac{B^2 + C^2 - A^2}{2BC}\right) \quad (56)$$

Once we calculate the inner angles of each triangle the given vertex is a part of, we can plug the values into the discrete curvature formula (4).

Thus the induced discrete curvatures for r is

Vertex e	Discrete Curvature $K_r(e)$ (up to 5 decimal places)
1	2.37781
2	4.59519
3	4.00207
4	4.73289

Thus the induced discrete curvatures for R is

Vertex e	Discrete Curvature $K_R(e)$ (up to 5 decimal places)
1	1.21223
2	5.21346
3	4.51403
4	4.76824

3.5. Verification of Counterexample.

By the definition of r and R , we see that $r|_B = (100, 100, 100) \leq R|_B = (110, 240, 220)$. Additionally, $K_r(4) \leq K_R(4)$, but $r(4) > R(4)$.

3.6. Conclusion of the Counterexample.

The premises hold ($r|_B \leq R|_B$, $K_r|_A \leq K_R|_A$) but $r(4) > R(4)$. This violates the conjecture on the disk. By a doubling argument, it also provides a counterexample for a closed surface.

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